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29 SEPTEMBER 1987
SUPERSEDING
MIL-T-7305B
4 JUNE 1964

MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, TORQUE, TYPE D-5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating torque pressure transmitter, designated D-5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-T-5350	Transmitter, Synchro Operated, Aircraft, General Specification For
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-I-7057	Indicator, Synchro, Aircraft, General Specifications For
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements For
MIL-I-7304	Indicator Pressure, Torque Sensitive, Type A-9

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-810C	Environmental Test Methods & Engineering Guidelines

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MS 3106	Connector, Plug, Electric, Straight
AN 816	Adapter, Straight, Pipe To Tube
AN 929	Cap Assembly, Tube Pressure Seal
AND 10050	Bosses, Standard Dimensions For Gasket Seal Straight Thread

(Copies of Specification and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the document which is DOD adopted shall be that listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issue of the document not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

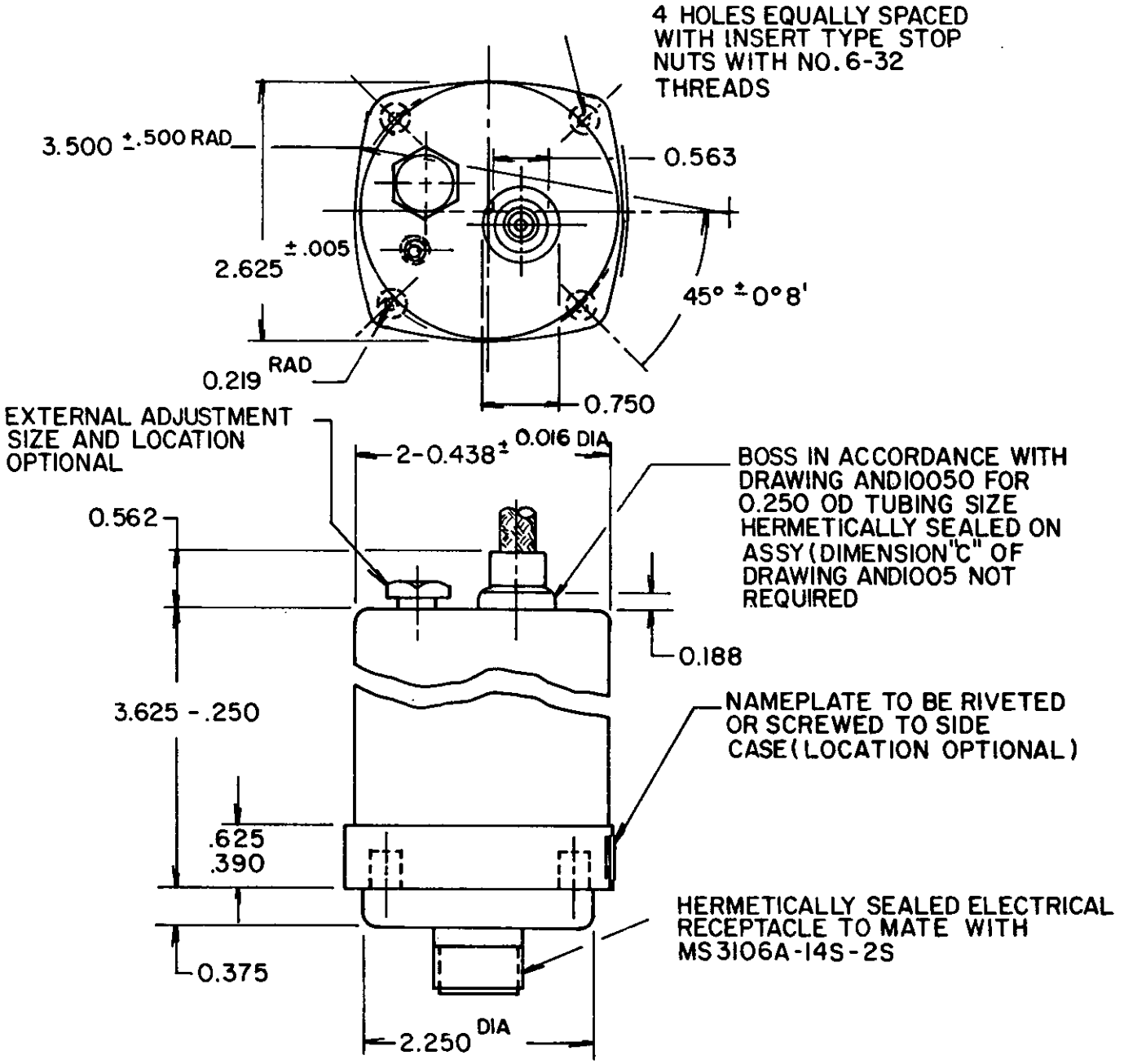
3.1 Qualification. Transmitters furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.1.1 Precedence. the requirements of MIL-T-5350 apply as requirements of this specification with the exceptions and additions specified herein. In the event of conflict this specification shall govern.

3.2 Reliability. The transmitter shall be so designed and constructed as to provide maximum practicable periods of mean time between failures (MTBF) of the transmitter. Failure shall be defined as any malfunction which results in degraded system performance, regardless of cause (careless handling or involuntary destruction excepted). The transmitter shall have a MTBFD of 750 hours.

3.3 Design and construction. The transmitter shall conform to figure 1. The transmitter shall be designed to operate with the type A-9 indicator, using the type A-13 amplifier conforming to MIL-I-7304.

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS 0.016

FIGURE 1. Transmitter.

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3.3.1 Transmitter assembly. The transmitter assembly shall consist of a transmitter in accordance with figure 1, a pressure fitting in accordance with figure 2, and a capillary link approved by the acquisition activity.

3.3.2 Synchro movement. For the range of 0 to 350 pounds per square inch (psi), the transmitter synchro shall move through an arc of 315 degrees. The angle of synchro movement shall be linear in relation to the pressure applied.

3.3.3 Case. The hermetically sealed case enclosing the transmitter element shall be equipped with a tear strip for convenience in disassembly. The end of the tear strip shall be plainly marked with a light gray line. The design of the case and tear strip shall provide for a reasonably simple means of resealing the case, and shall be of such nature that resealing can be accomplished with tools ordinarily possessed by an instrument repair man.

3.3.4 Reset adjustment. A hermetically sealed external zero reset adjustment shall be included on the end of the case containing the pressure connection as shown on figure 1. The adjustment shall have a total of the not less than 15 angular degrees movement of the transmitter synchro.

3.3.5 Pressure measuring unit. The pressure measuring unit of the transmitter shall include a barometric pressure compensating element vented to the atmosphere through a porous metal filter. The design shall provide a simple means of replacing the filters.

3.3.6 Marking. The top of the transmitter when in its normal operating position shall be clearly marked "TOP" with light gray lettering.

3.4 Mounting. The pressure fitting of the transmitter assembly shall be designed for direct mounting on the engine housing and shall conform to figures 2 and 3. The capillary system shall be substantially unaffected by exposure to temperature extremes, and shall be designed to eliminate oil congealing. The transmission capillary shall be an integral part of the system and shall not require servicing throughout the life of the transmitter. The capillary shall be 8 feet $\pm$ 0.500 inch long, and shall be unaffected by vibration encountered in engine nacelle installation. The capillary shall be legibly and permanently marked with a minimum of three bands containing the following statement:

CAUTION-DO NOT CUT CAPILLARY

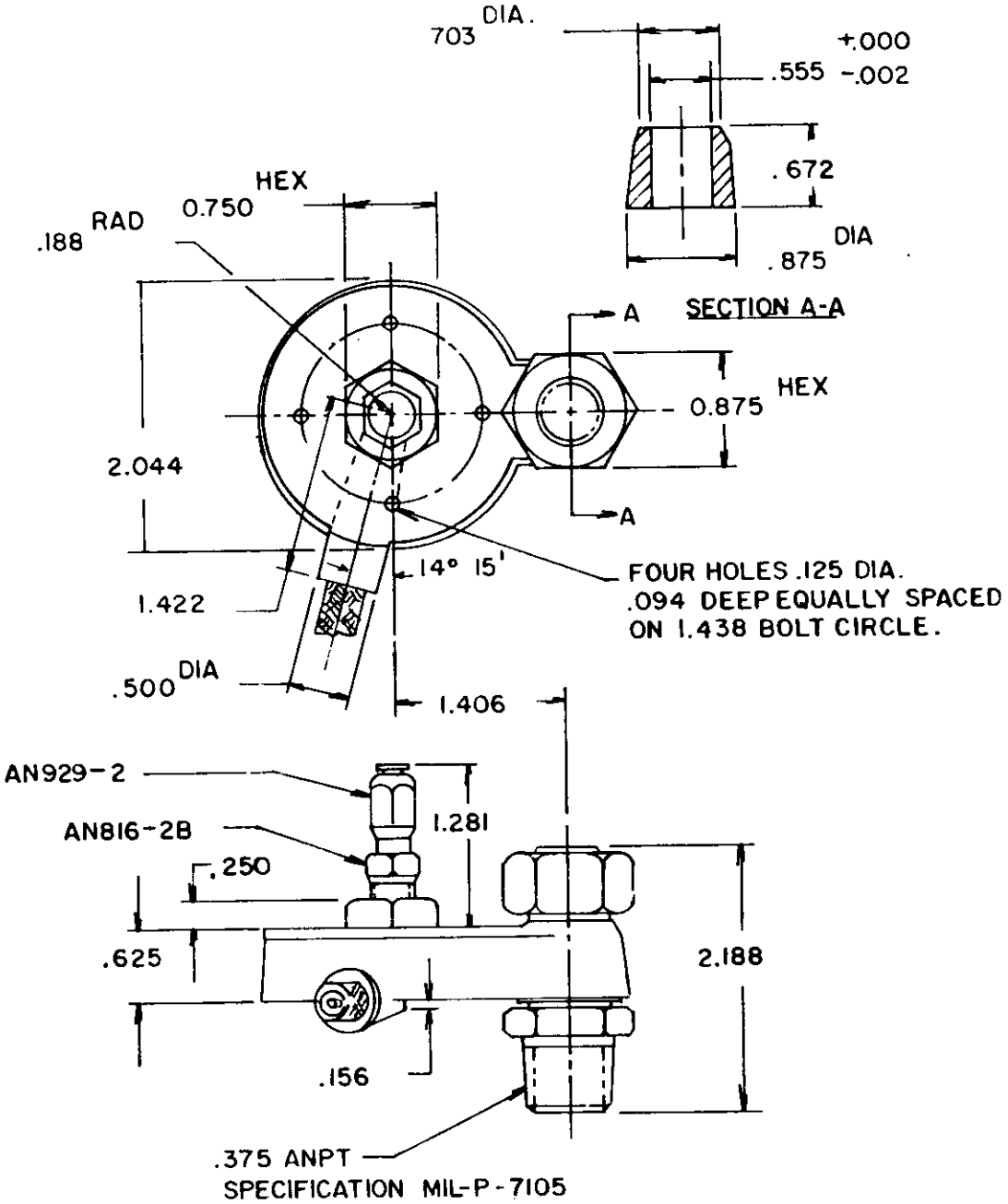
3.5 Damping. The transmitter assembly shall be sufficiently damped to prevent oscillation of the indicator pointer, but shall not introduce an objectionable lag in pointer response. The damping shall be accomplished as shown on figure 3, or as otherwise specified by the acquisition activity (see 6.2).

3.6 Locking. The pressure fitting over shall be locked with a suitable locking device.

3.7 Stop. A stop shall be provided, and shall be so designed as to restrain the prime mover. The stop shall allow the pointer of the indicator to travel beyond full scale to a point approximately midway between the last graduation and the 12 o'clock of the long pointer.

3.8 Weight. The weight of the complete torque pressure transmitter assembly shall not exceed 2.9 pounds.

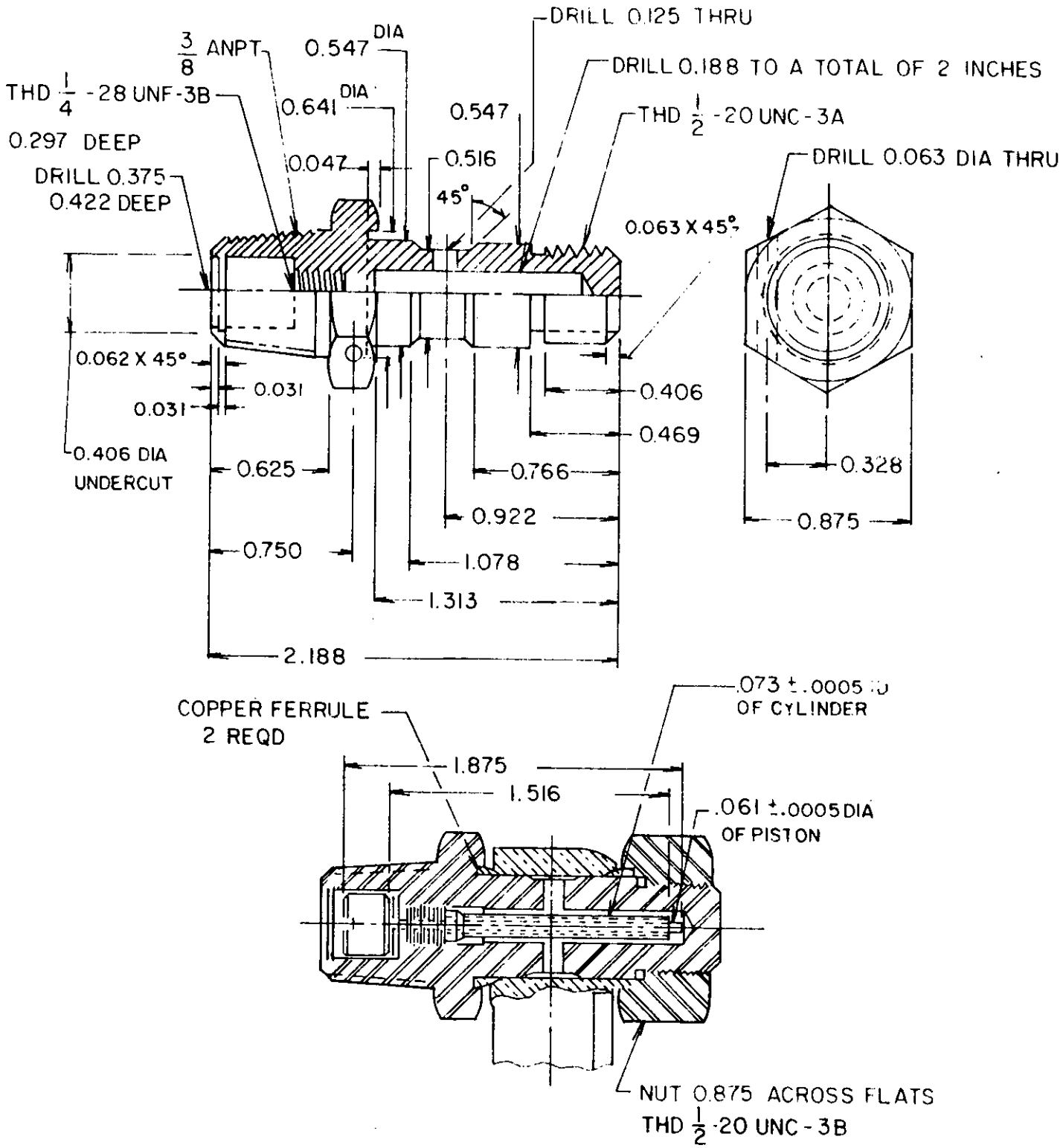
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCES: DECIMALS ± .016, ANGLES ± 1°.

FIGURE 2. Pressure fitting.

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES : FRACTIONS 0.01
ANGLES ±°

FIGURE 3. Damper.

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3.9 Pipe threads. Pipe threads shall be in accordance with MIL-P-7105.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.11 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The examining and inspection of the transmitter, pressure fitting, and capillary link shall be classified as:

(a) Qualification inspections (4.3)
(b) Quality conformance inspections (4.4)

4.3 Qualification inspections. Qualification inspections shall consist of all the examinations and inspections specified in 4.6 and 4.7.

4.3.1 Qualification samples. The qualifications inspection samples shall consist of three transmitters. Samples shall be identified with the manufacturer's part number and any additional information required by the letter of authorization (see 6.3).

4.4 Quality conformance inspections. Quality conformance inspections shall consist of:

(a) Individual inspections (4.4.1)
(b) Sampling inspections (4.4.2)

4.4.1 Individual inspections. Each transmitter shall be subjected to:

(a) Detail examination (4.6.1)
(b) Scale error at room temperature(4.7.1)
(c) Friction (4.7.2)
(d) Position error (4.7.3)
(e) Dielectric strength (4.7.4)
(f) Leak (4.7.5)

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4.4.2 Sampling inspections.

4.4.2.1 Sampling plan A. Five transmitters selected at random from each 500 or less on the contract or order shall be subjected to:

(a) Individual inspections (4.4.1)
(b) Transmitter assembly high (4.7.6) and low temperature
(c) Vibration error (4.7.7)
(d) Electrical characteristics (4.7.8)
(e) Pressure fitting high (4.7.9) temperature

4.4.2.2 Sampling plan B. Unless otherwise specified (see 6.2), five transmitters selected at random from the first 25 items of the contract or order shall be subjected to:

(a) Sampling plan A (4.4.2.1)
(b) Torque (4.7.10)
(c) Low temperature exposure (4.7.11)
(d) Altitude (4.7.12)
(e) High temperature exposure (4.7.13)
(f) Seasoning (4.7.14)
(g) Endurance (4.7.15)
(h) Radio noise suppression (4.7.16)
(i) Voltage-frequency variation (4.7.17)
(j) Salt spray (4.7.18)
(k) Overpressure (4.7.19)
(l) Damping (4.7.20)
(m) Capillary link vibration (4.7.21)

4.5 Inspection conditions. Inspection conditions shall be as specified in MIL-T-5350, and as follows.

4.5.1 Position. Unless otherwise specified, the transmitter shall be inspected with the longitudinal axis in a horizontal plane, and the pressure fitting and transmitter shall be approximately the same height to avoid head pressure errors.

4.6 Examinations.

4.6.1 Detail examination. The transmitter and pressure fitting shall be inspected to determine compliance with regard to workmanship, materials and marking.

4.6.2 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance with section 5.

4.7 Inspection methods.

4.7.1 Scale error at room temperature. The transmitter shall be inspected for scale error, first with increasing readings, then with decreasing readings at the inspection points specified in table I. With increasing readings, the indication shall be bought up to, but shall not exceed the indication specified to give the desired readings, and with readings decreasing, the indication shall be brought down to, but shall not fall below the indication specified to give the desired reading. Tolerances shall not exceed those specified in table I.

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TABLE I. Scale error inspection (room temp.).

Psi	Tolerance ± psi
0	3.0
50	2.5
100	2.5
120	1.0
140	1.0
160	1.0
180	1.0
200	1.0
220	2.5
240	2.5
300	3.0
350	3.5

4.7.2 Friction. The friction error inspection may be conducted in conjunction with the inspection specified in 4.7.1. Two readings shall be taken; the first before and the second after the transmitter is tapped. The difference between the first and second readings shall not exceed the tolerance specified for friction in table II.

TABLE II. Tolerance for other inspections.

Tests	Tolerance, ± psi
Friction error:	
Room temperature	1.5
Low temperature	3.0
High temperature	2.0
Position error at 200 psi	1.0
Altitude	2 times table I
Scale error:	
Low temperature	2 times table I
High temperature	2 times table I
Low temperature exposure	1.0 (+ table I)
High temperature exposure	1.0 (+ table I)
Seasoning	1.0 (+ table I)
Endurance	2 times table I
Vibration:	
Pointer oscillation	1.0
Scale error	0.5 (+ table I)
Overpressure	0.5 (+ table I)
Pressure fitting	
High temperature	0.5 (+ table I)
Transmitter assembly	
High and low temperature	2 times table I
Voltage-frequency variation	1.5 times table I

4.7.3 Position error. The position error inspection may be conducted in conjunction with the inspection specified in 4.7.1. The transmitter shall be energized to obtain a reading of approximately 200 psi. The change in indication produced by rotating the transmitter from the normal operating position to a

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position 90 degrees clockwise and counterclockwise (longitudinal axis vertical) shall be noted. The maximum change in indication shall not exceed the tolerance specified for position error in table II.

4.7.4 Dielectric strength. A potential of 200 volts (V) at a commercial frequency shall be applied between any pin of the electrical receptacle and the transmitter case for a minimum of 5 seconds. There shall be no breakdown of the insulation.

4.7.5 Leak. A pressure of 350 psi shall be applied to the pressure fitting, and a pressure of 14 psi shall be applied to the vent connection. The connecting tubes shall be completely sealed at a point within 2 inches of each connection. During a period of 1 minute, the pointer shall not change its position. This inspection may be combined with the inspection specified in 4.7.1.

4.7.6 Transmitter assembly high and low temperature. The pressure fitting shall be saturated for 4 hours at $90^{\circ} \pm 2^{\circ}$ C, and the major portion of the capillary line and the transmitter shall both be saturated at $-54^{\circ} \pm 2^{\circ}$ C for 4 hours. A scale error inspection shall be conducted with the components maintained at the above specified temperatures. The scale error shall not exceed the tolerance specified in table II.

4.7.7 Vibration error (shock absorbing mounted transmitters). The transmitter shall be subjected to vibration for 3 hours on a suitably designed vibration stand at an amplitude of 0.018 to 0.020 inch at a frequency range from 300 to 3,000 cycles per minute (cpm). During this period, a pressure shall be applied to the transmitter sufficient to produce approximately half scale deflection. The amplitude of pointer oscillation of a master indicator shall not exceed the tolerance specified in table II. The frequency shall be varied from 300 to 3,000 cpm, and observation shall be made for any resonant points. If the transmitter does not indicate any resonance within the frequency range, the frequency shall be set at any point between 2,000 and 3,000 cpm. If resonance is noted, the transmitter shall be vibrated at that point for the duration of the inspection. Following the vibration period, the transmitter shall be inspected for scale errors as specified in 4.7.1, on increasing readings only. The scale error shall not exceed the tolerances specified in table II.

4.7.8 Electrical characteristics. The transmitter shall be inspected for compliance with MIL-T-5350, for primary current, power input, secondary voltage, and for operating either one or two indicators without a change in indication of more than 1 degree of arc.

4.7.9 Pressure fitting high temperature. The pressure fitting shall be saturated at a temperature of $90^{\circ} \pm 2^{\circ}$ C for 4 hours. The capillary line, transmitter, and indicator shall be at room temperature. A scale error inspection shall be conducted with the pressure fitting at the high temperature. The scale error shall not exceed the tolerance specified in table II.

4.7.10 Torque. The rotor of the transmitter synchro shall be locked in one position and the pointer of the indicator synchro rotated through an angle from its corresponding synchronous position. The torque tending to return the pointer of the indicator synchro to its synchronous position shall be measured. The torque is directly proportional to the angular displacement within displacements up to 20 degrees and shall be no less than 2,000 milligram millimeters per degree of displacement.

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4.7.11 Low temperature exposure. The transmitter shall be placed within the inspection chamber and cooled to and maintained at a temperature of $-65^{\circ} \pm 2^{\circ}$ C for a period of 72 hours. Then the transmitter shall be checked for operation throughout its range while at this temperature. The temperature shall then be raised to $-55^{\circ} \pm 2^{\circ}$ C and maintained at that temperature for not less than 24 hours. While at this temperature the transmitter shall be subjected to and shall pass the scale error inspection specified in 4.7.6. The transmitter shall be returned to room temperature for a minimum of 4 hours and then shall be subjected to a scale error inspection at room temperature. The scale error at room temperature shall not exceed the tolerances specified for room temperature in Table II.

4.7.12 Altitude. The pressure fitting, capillary line, and transmitter shall be subjected to a pressure equivalent to 50,000 ± 500 feet. A scale error inspection shall be conduced while the assembly is at the specified altitude. The scale error shall not exceed the tolerance specified in table II.

4.7.13 High temperature exposure. The transmitter shall be placed in a chamber and subjected to a temperature of $70 \pm 2^{\circ}$ C for 50 hours. The transmitter shall be energized throughout the inspection. At the end of this period and with the transmitter still at this temperature the transmitter shall be subjected to a scale error and friction inspection. The scale error shall not exceed the tolerance specified in table II.

4.7.14 Seasoning. The transmitter shall be operated to indicate equivalent of zero indication to 80 ± 5 percent full scale and back to zero indication for 10,000 times at a rate of 30 ± 10 cpm. The transmitter shall be inspected not less than 1 hour after this inspection for scale error. The scale error shall not exceed the tolerance specified in table II. The transmitter shall be examined to determine if undue wear has occurred.

4.7.15 Endurance. The transmitter shall be subjected to 60,000 applications of pressure to produce an indication from zero to 80 ± 5 percent of the range of the transmitter and back to zero indication at a rate of 30 ± 10 cpm. The transmitter shall be inspected not less than 1 hour this inspections for scale errors at room temperature and the tolerances shall not exceed those specified in table II.

4.7.16 Radio noise suppression. Radio noise suppression shall be in accordance with MIL-I-6181, expect that a frequency range of from 0.15 to 150 megacycles shall apply for the radiated radio noise inspection.

4.7.17 Voltage-frequency variation. The transmitter shall be subjected to room temperature scale error inspections at the following voltages and frequencies:

(a) 23.5V at 380 cps

(b) 30.0V at 430 cps

The maximum difference in readings obtained from the room scale error inspection at 26V, 400 cps and the scale inspections at the voltages and frequencies shown in (a) and (b) herein shall not exceed the tolerances specified in table II.

4.7.18 Salt spray. The transmitter shall be subjected to a salt spray inspection in accordance with MIL-STD-810C, Method 509, for a period of 50 hours. The transmitter shall then be examined and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation. The transmitter

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shall be subjected to the inspection specified for dielectric strength and the measured resistance shall be a minimum of 20 megohms. The external surface of the transmitter may be washed with distilled water and air dried prior to its subjection to the dielectric strength inspection.

4.7.19 Overpressure. The transmitter shall be subjected to an applied pressure of 600 psi for 10 minutes. The transmitter shall be inspected not less than 1 hour following the overpressure inspection for scale error as specified in 4.7.1. The scale error shall not exceed the tolerance specified in table II.

4.7.20 Damping. A synchro indicator in accordance with MIL-I-7057 shall be electrically connected to the transmitter. A pressure of 350 psi shall be applied to the transmitter assembly. A quick-release valve shall be used to exhaust the pressure. The time lapse for the indicator pointer to travel 300 circular degrees shall be not less than 1 second nor more than 3 seconds. The oil used for this inspection shall be SAE 10, or an oil having an equivalent viscosity at room temperature.

4.7.21 Capillary link vibration. The pressure transmission link shall be subjected to a vibration inspection as described in 4.7.21.1 through 4.7.21.5 by securely mounting the pressure line and fitting to the vibration plate of the vibration stand. Seven inches from the pressure fittings the line shall be securely clamped to a stationary bracket. The flexible section of the line contained between the two brackets shall have no bend in it. After completion of the inspection, the flexible line shall be vibrated along three mutually perpendicular axes for a total of 100 hours, at an amplitude of 0.030 to 0.040 inch and at a frequency of 2,600 to 3,000 cpm. Throughout the vibration inspection, a pressure of 200 psi shall be applied to the pressure line. The transmission line shall be examined for damage and there shall be no evidence of failure.

4.7.21.1 Capillary link vibration inspection procedure. This procedure applies to equipment assemblies (including any resilient mounting subassemblies) which mount directly on gasturbine engines or on the structure of aircraft powered by reciprocating, turbojet or turbo-propeller engines. This procedure also applies to equipment installed in missiles powered by turbojet engines. This procedure represents a standard of performance. More severe requirements in regard to frequency range and acceleration level than those herein stipulated may be necessary where equipment is to be installed in extreme environments such as might exist, for example, near intense jet and assist rocket engine exhaust noise sources and on fuselage sidewalls in propeller plane areas. Equipment which meets this standard is generally suitable for:

a. Installation in existing types of aircraft.
b. Installation in most locations on the aircraft structure.
c. Applications on new aircraft.

4.7.21.2 Inspection specimen. The specimen shall be attached to a rigid fixture capable of transmitting the vibration conditions specified herein. Attachment of the specimen to the fixture shall be that which represents dynamically the most adverse of alternate service mountings. The inspection specimen shall be functioning and shall remain within the operational requirements in accordance with the provisions of the detail specification during the entire inspection. If the operational environment exceeds 71° C (106°F), half of each resonant and cycling period shall be conducted at room temperature and half at the applicable high temperature as specified by the contracting activity. Inspections shall be

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conducted under both the resonant and cycling conditions specified herein, the order being optional. The amplitude of applied vibration shall be monitored on the inspection fixture near the specimen mounting points.

4.7.21.3 Resonance. Resonance modes of the inspection specimen shall be determined by varying the frequency of applied vibration slowly through the specified range at vibratory accelerations not exceeding those shown in figure 4, curve A. Individual resonance surveys shall be conducted with vibration applied along each axis of any set of three mutually perpendicular axes of the inspection specimen. The specimen shall be vibrated at the indicated resonant conditions for the periods shown in the Vibration inspection schedule and with the applied double amplitudes of vibratory accelerations in figure 4, curve A. The vibratory acceleration applied to specimens weighing more than 50 pounds may be decreased by ±1g for each 10 pound increment of weight above 50 pounds to a value no lower than ±5g, if so approved by the contracting activity. These periods of vibration shall be accomplished with vibration applied along each of the three mutually perpendicular axes of vibration. When more than one resonance is encountered with vibration applied along any one axis, each resonance shall be sustained for the period shown in the applicable portion of the vibration inspection schedule. If more than four resonances are encountered with vibration applied along any one axis, the four most severe resonances shall be chosen for inspection.

4.7.21.4 Cycling. The specimen shall be vibrated under the cycling conditions specified herein for the applicable periods listed in the vibration inspection schedule (table III). The frequency shall be cycled between 5 and 500 cps at an applied double amplitude of 0.036 inch or an applied acceleration of ±10g whichever is the lower value. The rate of change of frequency shall be logarithmic, and such that 15 minutes are required to proceed from 5 to 500 back to 5 cps. When there is no provision for logarithmic cycling, a linear rate of frequency change may be used.

TABLE III. Vibration inspection schedule.

(Times shown refer to one axis of vibration)					
Number of resonances	0	1	2	3	4
Total vibration time at resonance	-	30 min	1 hr	1.50 hr	2 hr
Cycling time	3 hr	2.50 hr	2 hr	1.50 hr	1 hr
* 30 minutes at each resonance					

4.7.21.5 Supplementary inspection electronic assemblies. In addition to the resonance and cycling inspections, electronic assemblies shall be removed from their external resilient mounting, rigidly installed on the vibrator, and subjected to the following inspections to demonstrate a minimum structural resistance to vibration. Vibration frequency shall be varied between 5 and 500 cps at a double amplitude of 0.010 inch or an applied vibratory acceleration of ±2g whichever is the lower value. The rate of frequency cycling shall be logarithmic and such that 15 minutes are required to proceed from 5 to 500 and back to cps. When there is no provision for logarithmic cycling, a linear rate of frequency change may be used. The inspection shall be sustained for two complete cycles along each of three

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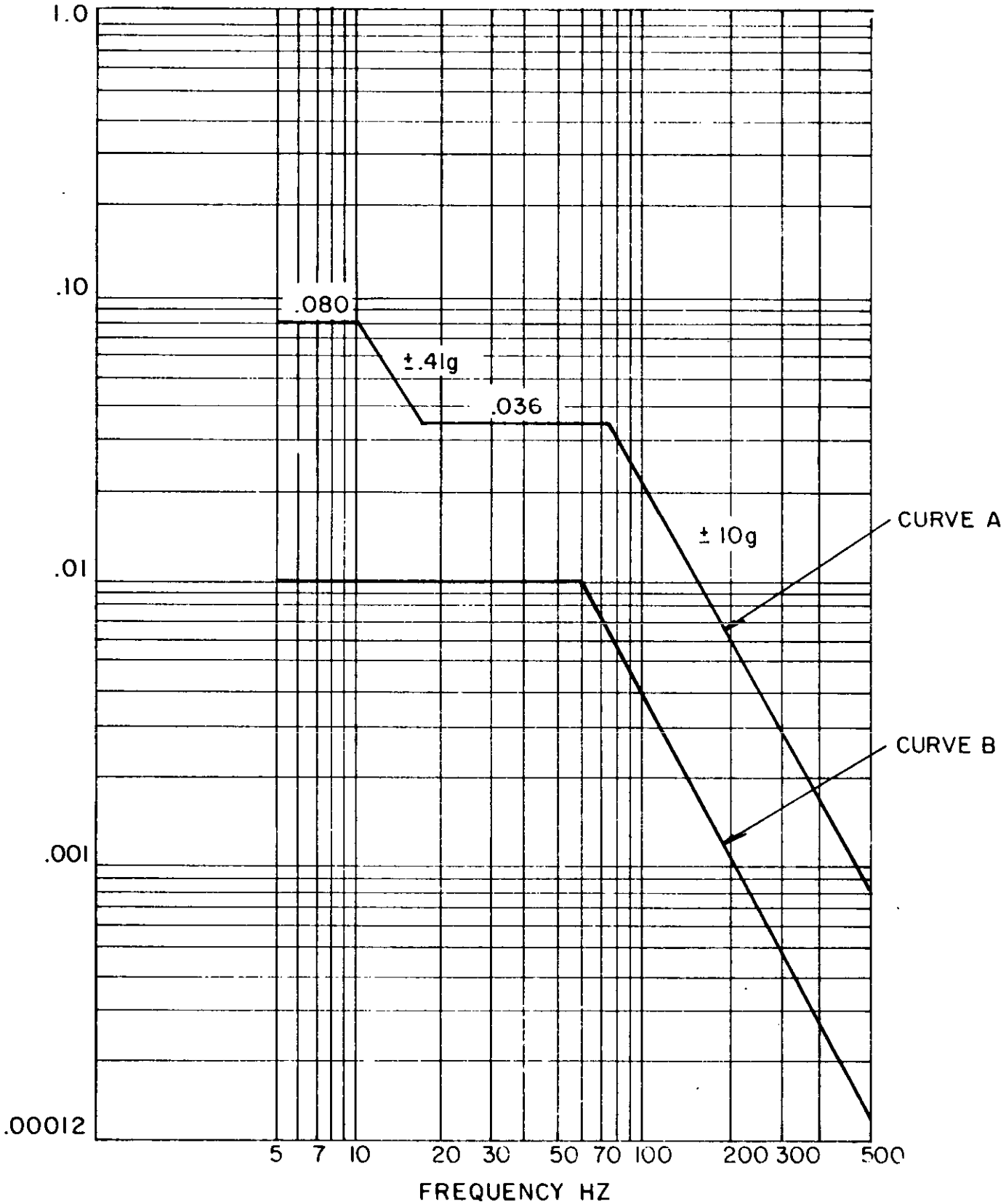


FIGURE 4. Range curves for vibration tests.

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mutually perpendicular axes. All resonance points shall be noted. In addition to this cycling inspection, each resonant condition shall be sustained for 10 minutes at the vibratory double amplitude or acceleration shown on figure 4, curve B.

5. PACKAGING

5.1. Preservation. Preservation shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1. Level A.

5.1.1.1. Cleaning. Transmitter should be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2. Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3. Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each tansmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3. Industrial. The Industrial preservation of transmitter shall be in accordance with ASTM D3951.

5.2. Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1. Level A. Transmitter packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2. Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1. apply.

5.2.3. Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4. Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

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5.3. Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1. Intended use. The type D-5 transmitter is intended for use as a transmitter component of a torque pressure indicating system on reciprocating engine aircraft.

6.2. Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Method of damping, if other than as specified (see 3.6).
(c) Applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Marking of shipments desired (see 5.3).
(e) Whether or not sampling plan B is to be omitted (see 4.4.2.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL-7305) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/MMEDO, Tinker AFB, OK 73145-5990, and information pertaining to qualification of products may be obtained from that activity,

6.4 Subject term (key word) listing.

Military specification
Transmitter, Pressure, Torque, Type D-5

6.5 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99
Navy - AS

Preparing Activity:
Air Force - 71

Project Number:
6620-0416

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-7305C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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